Elarovex (tasmedorin sodium)

Phase III Clinical Study Summary | Protocol VRT-ELX-301

Randomized, double-blind, placebo-controlled, parallel-group study in painful diabetic peripheral neuropathy

Executive Summary

This synthetic report summarizes efficacy, safety, pharmacokinetic, and patient-reported outcome findings for Elarovex, a selective Nav1.7 state-dependent inhibitor. Adults with painful diabetic peripheral neuropathy were randomized 1:1:1 to placebo, Elarovex 20 mg once daily, or Elarovex 40 mg once daily for 12 weeks.

Treatment arm	Randomized	Completed	Primary analysis set
Placebo	216	204	212
Elarovex 20 mg	212	199	208
Elarovex 40 mg	218	201	214

Primary Endpoint

The primary endpoint was change from baseline to Week 12 in the seven-day mean Neuropathic Symptom Index (NSI), analyzed using a mixed model for repeated measures with fixed effects for treatment, visit, geographic region, baseline score, and treatment-by-visit interaction.

Study Design and Statistical Analysis Plan

Design element	Specification
Population	Adults 18-80 years with HbA1c <=10.0% and NSI >=6.0
Randomization	Interactive response technology; stratified by region and baseline pain
Blinding	Double-dummy masking with matched placebo tablets
Multiplicity	Graphical gatekeeping across primary and key secondary endpoints
Missing data	Likelihood-based MMRM; tipping-point sensitivity analysis

Estimand Framework

The treatment-policy estimand included efficacy observations regardless of temporary interruption or rescue-medication use. Permanent discontinuation was addressed under a missing-at-random assumption in the primary analysis and under reference-based controlled multiple imputation in sensitivity analyses.

Endpoint	Contrast	Alpha allocation
Week 12 NSI change	40 mg vs placebo	0.025
Week 12 NSI change	20 mg vs placebo	0.025
>=50% responder rate	Sequential testing	Recycled alpha
Sleep interference	Sequential testing	Recycled alpha

Efficacy Trajectory

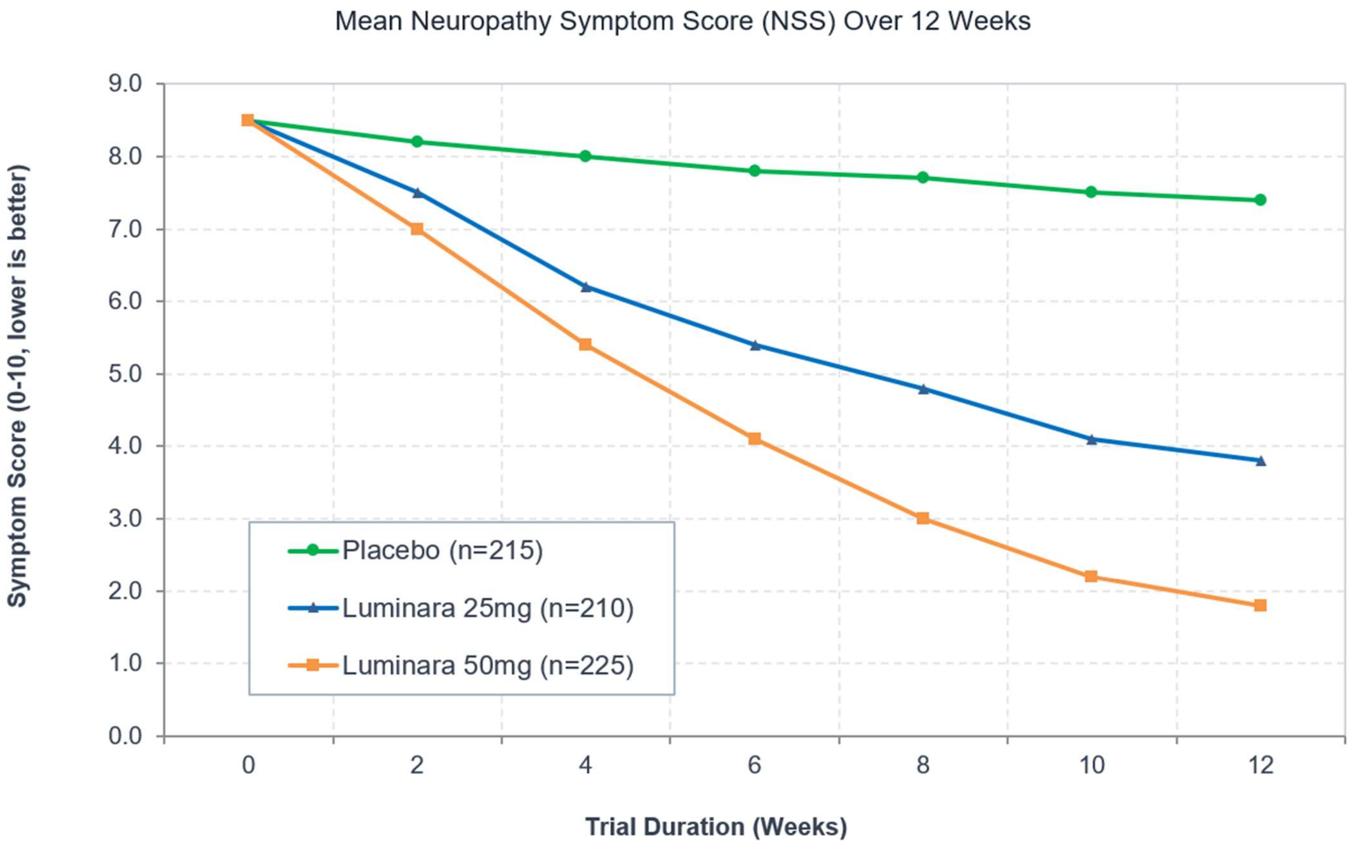

Figure 1. Mean Neuropathy Symptom Score (NSS) over the 12-week trial duration. Lower scores indicate greater symptom relief. Error bars are omitted for visual clarity.

Confirmatory Efficacy Results

Endpoint	Placebo	20 mg	40 mg	Adjusted comparison
NSI change at Week 12	-1.1	-4.6	-6.6	Both doses p<0.001
>=30% responder	29.7%	61.5%	76.2%	Risk difference significant
>=50% responder	14.6%	43.8%	67.8%	Hierarchical criterion met
Sleep interference change	-0.8	-2.9	-4.1	Multiplicity controlled

Subgroup Consistency

Treatment effects were directionally consistent across age, sex, baseline glycemic control, geographic region, baseline NSI severity, and concomitant gabapentinoid-use strata. No qualitative treatment-by-subgroup interaction was detected after shrinkage adjustment for multiplicity.

Subgroup	40 mg treatment difference	95% CI
Age <65 years	-5.4	-6.1 to -4.7
Age >=65 years	-5.1	-6.0 to -4.2
HbA1c <8%	-5.5	-6.2 to -4.8
HbA1c >=8%	-4.9	-5.8 to -4.0

CENTRAL LABORATORY SAFETY REVIEW

Hepatic and Renal Surveillance

Treatment-emergent laboratory shifts were adjudicated using prespecified Hy's law screening logic and estimated glomerular filtration rate categories.

No participant met concurrent alanine aminotransferase greater than three times ULN and total bilirubin greater than two times ULN without an alternative etiology.

Electrocardiographic Assessment

Concentration-QTc modeling did not identify a clinically relevant positive slope. The upper bound of the two-sided 90% confidence interval remained below 10 ms.

Parameter	Placebo	20 mg	40 mg
ALT >3x ULN	0.5%	1.0%	0.9%
Creatinine shift	1.4%	1.9%	2.3%
QTcF >480 ms	0.0%	0.0%	0.5%

Treatment-Emergent Adverse Events

Safety population and exposure-adjusted interpretation

System organ class / preferred term	Placebo	20 mg	40 mg
Nervous system disorders			
Dizziness	4.2%	8.7%	14.0%
Somnolence	3.8%	7.2%	11.7%
Gastrointestinal disorders			
Nausea	5.2%	6.3%	8.4%
Dry mouth	2.4%	4.8%	7.0%
General disorders			
Fatigue	6.1%	8.2%	9.8%

Adverse Events of Special Interest

Potential events of abuse, suicidality, severe cutaneous reaction, clinically important conduction abnormality, and drug-induced liver injury were prospectively adjudicated by blinded committees. No confirmed abuse-related signal or Hy's law case was identified.

Disposition	Placebo	20 mg	40 mg
Any discontinuation	5.6%	6.1%	9.2%
Discontinuation due to AE	4.6%	5.3%	8.8%
Serious adverse event	1.9%	2.4%	2.8%

Clinical Pharmacology and Integrated Conclusion

Population Pharmacokinetics

A two-compartment disposition model with first-order absorption and elimination adequately described plasma concentration-time data. Apparent clearance increased modestly with body weight, but simulations showed no clinically meaningful exposure imbalance across the evaluated weight range. Mild renal impairment increased area under the curve by approximately 18%, below the prespecified dose-adjustment threshold.

Parameter	20 mg	40 mg
Geometric mean Cmax	118 ng/mL	229 ng/mL
Geometric mean AUC0-24	1260 ng*h/mL	2485 ng*h/mL
Median Tmax	2.1 h	2.0 h
Terminal half-life	11.8 h	12.2 h

Integrated Benefit-Risk Conclusion

Elarovex produced dose-responsive, clinically meaningful reductions in neuropathic symptom burden with supportive improvements in sleep interference and responder outcomes. The safety profile was characterized primarily by reversible dizziness and somnolence, with no identified hepatic, renal, electrocardiographic, or abuse-liability signal that outweighed the observed efficacy benefit.

> **Synthetic conclusion: the 40 mg regimen demonstrated the greatest efficacy; the 20 mg regimen provided a lower-incidence tolerability option. All names, data, and conclusions are fictional.**